

Lecture 7

Writing Reports

Reports vs. Progress Updates

- Audience
 - Progress updates are typically for people immediately involved
 - Reports get shared – future team members, other interesting parties, etc.
- Timeliness
 - Progress updates are consumed immediately
 - Reports may be read years later
- Scope
 - Progress updates summarize current actions
 - Reports include all actions
- Purpose
 - Progress updates are usually to *inform*
 - Reports are usually to *document* and to *persuade*

Audiences for Reports

- Primary audience
 - Project sponsor
 - Other interested stakeholders
- Secondary audience
 - Yourself, in the future
 - The project sponsor, also in the future
 - Other team members who want to build on your work
- Unintended audience
 - People who get the report shared with them later on

Elements of a Report

- Remember the structures of an argument
 - Introduction -> Background -> Proposition -> Proof (Classical Method)
 - Claim -> Evidence -> Warrant (Toulmin Method)
- Reports can be structured the same way!
 - Introduction
 - Classical arguments, this is an overview of the problem and actions taken
 - Toulmin arguments, this also includes the key recommendations
 - Background
 - Methods
 - Results
 - Conclusions and Recommendations

Report Length

- Report length can vary a lot depending on the context and medium
 - Written report at the end of a long project – long!
 - Presentation at an executive meeting – short!
- Projects will often have *more than one* report
 - A blog post explaining your methods
 - A written report for stakeholders
 - A presentation to the project sponsor
- As reports get shorter, remove details and simplify

Methods

- DO include
 - Everything you will be presenting the results of
 - A description of the data you used
 - Explanations that are relevant to your audience
- DON'T include
 - Code!
 - Things that you won't present results for

Results – Example 1

- This example is from a classification project where we chained three different classification models, each using different features
 - If one model did not predict a class with high confidence, it proceeded to the next model
- This is one of a few similar tables presenting results for the same approach on different test sets

Model	True Positives	Misclassifications	Skips	Docs Processed
Model A	482	33	423	938
Model B	39	26	358	423
Model C	91	11	256	358
OVERALL	612 (65.2%)	70 (7.5%)	256 (27.3%)	938

Results – Example 2

- This example is from a recommendation engine project that compared different approaches
- This is one of several similar box plots that showed different metrics (*e.g.* Brier score, coverage error)

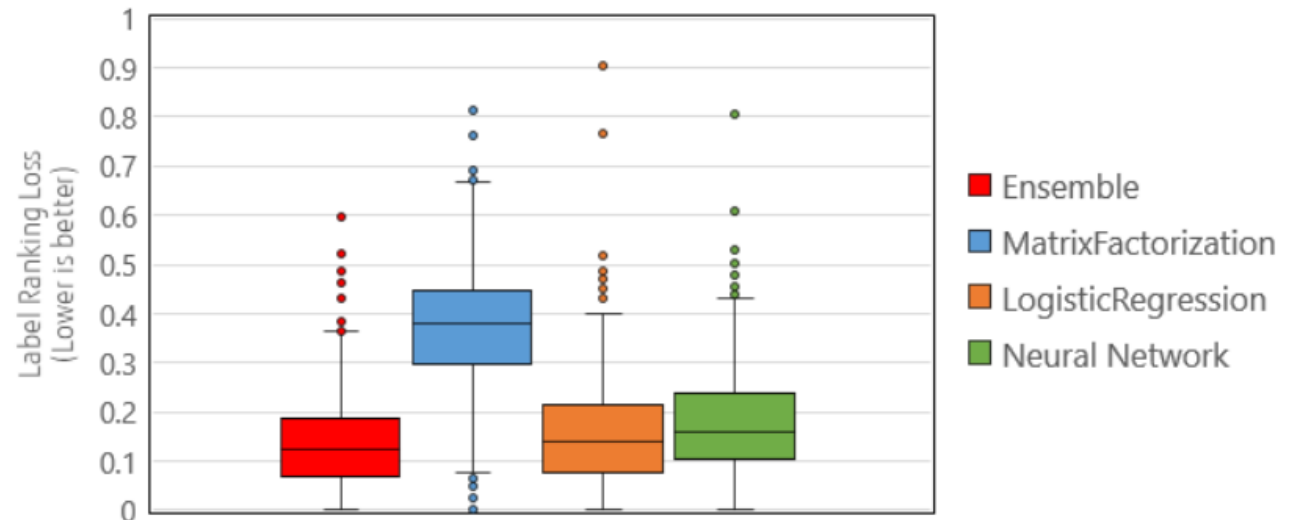
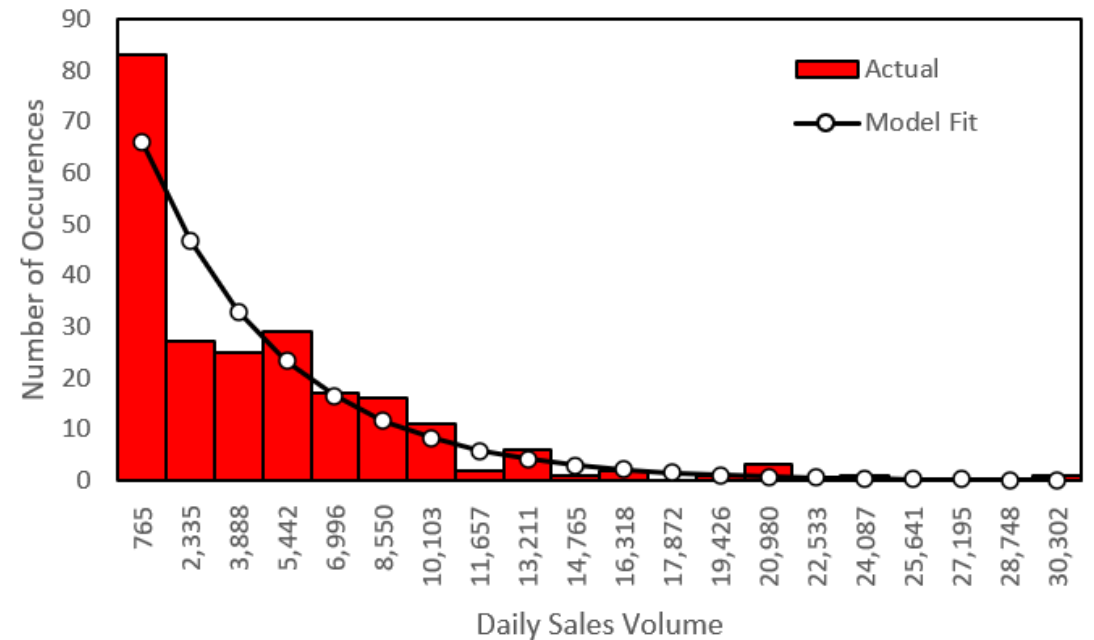


Figure 11: Box-and-whisker plot of the label ranking loss for each model

Results – Example 3

- This example is from a sales prediction model for a wholesaler
- The model wasn't used directly but fed into a safety stock prediction model
 - Safety stock is the inventory level at which a product is re-ordered so that the new order arrives before the remaining stock is used



Recommendations

- Recommendations are *proposals for future work*
 - Decisions to be made based on your findings
 - Further analysis on a related topic
 - Deployment of the solution in a workflow
- Recommendations *can* include ways to make your work better
 - Additional analysis
 - Collecting new data
 - Trying new methods

Common Errors

- Too much data, or data that is too dense
- Not providing clear outcomes or recommendations
- Poor formatting
 - Inconsistent fonts, colours, etc.
 - Text that's too small in figures
 - Missing or vague axis labels